

9618 CAIE Computer Science — Security and Integrity

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1 Terminology

Phishing, Pharming, Malware (Virus, Worm, Trojan Horse, Logic Bomb, Spyware, Bots)

Phishing: trying to get the user to click on a bad link.

Pharming: Uses **DNS Poisoning** to redirect your page request to a bad site.

Data Security: Ensures that data are not going to be lost or corrupted. (Frequent backups)

Data Privacy: Prevents unauthorised access (hackers) from accessing the data. This is achieved with firewalls, antivirus softwares etc.

Data Integrity: Ensures that data represents what it's supposed to represent. We use Validation and Verification for that.

2 Validation

This is the test to ensure that data are entered in the correct format. There are many different ways for this.

Type, range, limit, format, existence, presence, length, consistency, uniqueness checks.

These are pretty self explanatory. Presence check: if the data exists where it should be. Existence: checks if the data in a database/file physically exists.

3 Verification

Ensures that data entered has integrity (accurately represents what it's supposed to) Examples are: **Double Entry, Visual Check, Checksum.**

The only deep one is the check sum:

3.1 Check Sum

There are also different types: we may use a check digit (at the end of a long list of numbers), and that is used with mod. Example usage: ISBN, Vehicle identification numbers, barcodes.

Parity Checks: Appends a parity bit at the end of the byte to make it up to the even/odd number of 1s. A **parity block** is just a table of bytes. You do the parity bit horizontally and vertically. Then you can identify where it is that you went wrong.

If the digits got interchanged though, the parity block doesn't work.